HOMEWORK ASSIGNMENT

Build and Benchmark Your Own MI-EEG Classification Pipeline

- In this assignment, you will build your own motor imagery EEG (MI-EEG) classification pipeline. You will design both an EEG preprocessing/feature-extraction pipeline and a neural network, integrate them into the existing repository, and compare your method with **MixNet**.

- To make the comparison fair, your method and MixNet must be evaluated under exactly the same experimental protocol: the same 9 subjects, the same 5-fold cross-validation procedure, the same test session, and the same evaluation metrics.

WHY THIS ASSIGNMENT

Learning Objectives

By the end of this assignment, you will be able to:

1. Develop and justify an original MI-EEG pipeline

Formulate a clear hypothesis, then design a preprocessing / feature-extraction scheme and network architecture that follows from it.

2. Conduct a fair, reproducible benchmark

Compare your method with MixNet under an identical experimental protocol, while preventing data leakage and using validation data correctly.

3. Analyze and communicate findings

Interpret per-subject and statistical results, assess whether they support your hypothesis, and discuss limitations and future directions.

THE EXPERIMENTAL GROUND TRUTH

Dataset and Experimental Protocol

BCI Competition IV-2a (BCIC2a)

Task	2-class motor imagery — left vs. right hand
Subjects	9 (A01–A09), evaluated subject-dependently
Sessions	T (train/validation) and E (test)
Validation	5-fold stratified cross-validation within session T
Metrics	Test accuracy and F1-score, per fold and per subject

THE RULE

Session E is an untouched test set.

It is never used for training a model and never used for choosing hyperparameters. Every fold is validated only within session T; session E is the test set, at the very end, for every fold.

WHAT YOU WILL BUILD

Assignment Structure — Three Tasks

TASK 1

Preprocessing & Feature Extraction

Design a preprocessing and feature-extraction pipeline that transforms raw EEG into the representation used by your network.

Complete:

`mixnet/preprocessing/BCIC2a/HW_prep.py`
- `fit_transform()`
- `transform()`

TASK 2

Neural Network Design

Design and implement a neural network motivated by your hypothesis, while following the shared training and evaluation framework used in the repository.

Complete:

`mixnet/models/HWNet.py`
- `build()`
- `_config()`

Tune the model and training hyperparameters in:
`experiments/configs/HWNet.py`

TASK 3

Benchmark & Report

Compare your method with the MixNet baseline under the same experimental protocol.

Run `benchmark.py` to:
- aggregate results across subjects
- compare per-subject performance

Presentation: 15 min-presentation

Don't worry about building the pipeline from scratch!!

You only need to complete the provided functions.
Setup instructions, environment configuration, and file-by-file implementation details are available in INSTRUCTION.md in the repository:
https://github.com/xydxdy/MixNet/tree/homework

If you have any questions, feel free to contact: **rattanaphon.c_s18@vistec.ac.th**

HOW THE WORK IS SUBMITTED AND JUDGED

Deliverables and Evaluation

Deliverables

- **Code:** your preprocessing module, model, and configuration files.
- **Report:** 15 min-presentation

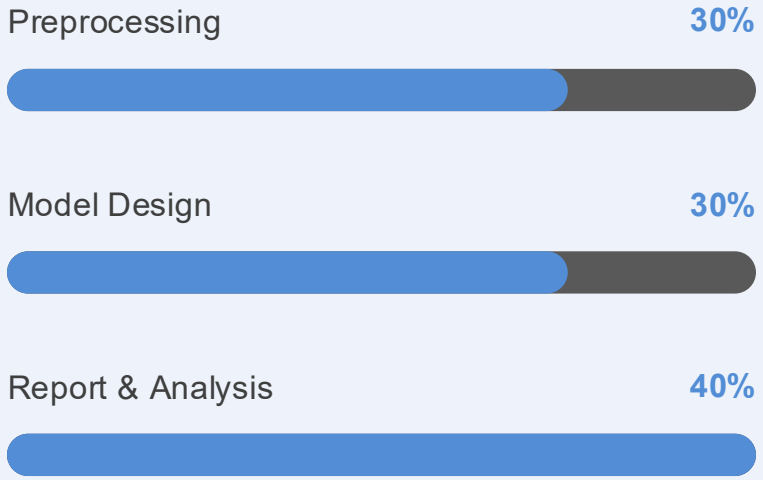
Key Rules

- Fit every data-driven step on the training fold only
  no leakage across folds or into session E.
- Use all 9 subjects, with 5-fold cross-validation, so the MixNet comparison stays paired.
- Cite any external code, repository, or paper you draw on.

Submission

- Email your **GitHub repository link** to **theerawit.w@vistec.ac.th** and **rattanaphon.c_s18@vistec.ac.th**

GRADING WEIGHTS



Outperforming MixNet is not required, but your method should achieve reasonably competitive performance.

Strong experimental design, a fair comparison, and well-supported analysis will earn the strongest grade.